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# Academic early warning model based on machine learning and model application

Qiang Li[1,2], Yihan Liu[1]✉, Rui Ma[1] & Qike Wu[1]

**With the expansion of higher education, the uncertainty of students' academic completion and the diversity of academic crises have posed new challenges to the management of higher education. This study aims to design and implement a dynamic academic early warning system based on machine learning to predict and intervene in students' academic crises. By analyzing the causes of academic crisis, the fuzzy comprehensive evaluation with analytic hierarchy process method is used to construct an academic early warning indicator system comprising 10 key indicators. This ensures the scientificity and rationality of the indicator system through expert scoring and consistency test. On this basis, a radial basis function neural network was used to construct an academic early warning model, which outperforms the recurrent neural network and Softmax regression model in terms of prediction accuracy and convergence speed. The system was developed using hypertext markup language, cascading style sheets, JavaScript, and Python to achieve a user-friendly human–computer interaction interface and provide personalized academic alert services. The experimental results demonstrate that the system exhibits high sensitivity and accurate recognition capabilities when dealing with large-scale student datasets, achieving an accuracy rate of 96.32% and a root mean square error of 0.2926, which meets the practical requirements of academic early warning. The results of this study not only provide a new academic early warning tool for colleges, but also have important practical value for promoting the construction of smart campus and digital campus. The current findings, drawn from a sample limited to engineering with an overrepresentation of males, require future validation in multi-disciplinary and gender-balanced cohorts to establish broader applicability.**

**Keywords** Academic early warning, Machine learning, Radial basis function, Neural network, Educational management, FCE-AHP

Currently, there is a growing concern regarding the increasing severity of issues such as grade repetition, school dropout rates, loss of eligibility to pursue desired majors, and even student dismissals[1,2]. Ensuring the effective culmination of students' academic pursuits has emerged as a pressing concern in the realm of higher education that requires immediate resolution[3,4]. Receiving warnings will heighten students' awareness of their current learning level, prompting them to modify their learning attitudes more effectively, exert greater effort to enhance their grades, and prevent academic probation. Additionally, the academic adviser possesses a partial understanding of the student's academic probation risk, enabling them to offer more suitable and prompt support to students[5].

Academic warnings for students allow the university to promptly address students' learning difficulties. Advisors can implement diverse strategies to aid students in enhancing their academic achievement or avoiding delayed graduation. The university's research project involves constructing an academic early warning model for undergraduate students using machine learning techniques. Consequently, an academic early warning system will be developed based on this model[6]. The primary purpose of the early warning system is to offer decision support to university management by accurately forecasting the academic risks faced by students. This is done in order to effectively protect and maintain the academic progress of students[7]. Consequently, an increasing number of scholars acknowledge the significant societal capacity of educational data and engage in studies in this field. For instance, the utilization of deep learning technology in academic early warning can proactively identify students who are at risk of repeating or dropping out of a course due to the accumulation of failed credits[8]. Furthermore, the implementation of this technology also facilitates the development of digital and

[1]College of Engineering, Hebei Normal University, Shijiazhuang, China. [2]Vocational and Technical College and Hebei Provincial Key Laboratory of Information Fusion and Intelligent Control, Hebei Normal University, Shijiazhuang, China. ✉email: liuyihansongjiawei@gmail.com

intelligent campuses in higher education institutions[9]. Despite the increasing use of the academic early warning system in colleges and universities and its demonstrated effectiveness, there are still some significant challenges in its implementation.

During the construction of the academic early warning index system, researchers encounter a multitude of intricate academic influencing aspects[10]. The feature characteristics that capture the pupils' academic process lack comprehensiveness. Feature generation is a notable technique that finds use in various machine learning and deep learning challenges[11]. Mingyu et al.[12] employed a categorization of attributes related to students' academic pursuits, housing conditions, online activities, and fundamental information to address the aforementioned issue. The Catboost-SHAP technique had the highest performance with an R-squared ($R^2$) value of 80.3%, a mean squared error (MSE) of 24.976, and a mean absolute error (MAE) of 3.551. Hamim et al.[13] categorized students' learning outcomes in 2022 according to many factors, including personal behavior, social behavior, study habits, family background, academic majors, and grades. Furthermore, they employed feature selection approaches to enhance the performance of the model. The experimental findings indicate that LightGBM is the most potent algorithm, with an accuracy of 89.26%. This performance was achieved by selecting 14 features using a feature selection technique. In their study, Schneider et al.[14] identified 105 elements that have an impact on academic achievement. These factors encompass various aspects such as students' personal circumstances, instructional methods, and external influences. Nevertheless, there are several reasons that fail to accurately recognize and intervene with students who are experiencing a crisis during their academic journey. Romero et al.[15] forecasted patterns in students' ultimate learning difficulties by analyzing data such as the quantity of information shared, volume of emails exchanged, number of tests finished, and student engagement on an online learning platform. Nevertheless, their approach solely relied on learning process data while disregarding the static fundamental information of individual students.

Additionally, numerous machine learning techniques, such as support vector machine, KNN, decision tree, and linear regression, are utilized to construct prediction models. In their study, Migueis et al.[16] gathered data from a sample of 2459 students in order to develop predictive models for students' academic success. The authors advocated using the random forest model, which achieved an accuracy rate of over 95% in predicting academic outcomes at an early stage of the students' educational journey. In a preliminary study conducted by Rimal et al.[17], the aim was to determine the most effective machine learning model for hyper-parameter tuning. The study focused on employing single, multiple, and deep neural networks to predict the grades of engineering students from Pokhara University, Nepal in 2019. Selamat et al.[18] put up two methodologies: one involves machine learning prediction using computational models, while the other entails visual analytics. The decision tree algorithm, random tree, random forest, and REP tree were employed to examine a computer systems design course for a total of 489 students from 2016 to 2019. The findings indicated that the J48 algorithm achieved a precision rate of 99.8%. Bujang et al.[19] introduced a probabilistic grade prediction model for SPOC with a limited number of samples, utilizing a correlation vector machine (RVM). The simulation results demonstrated that RVM exhibits superior forecast accuracy and offers technical assistance in the distribution of SPOC. Nevertheless, the utilization of machine learning fails to adequately prioritize the examination of the link among factors.

Furthermore, as research and automation continue to advance, numerous intricate challenges arise. When algorithms are the only method used, many aspects of the data are not completely utilized. The use of educational data to forecast pupils' academic achievement has become a prominent area of research. A.J. Lekan conducted a study to examine the correlation between students' academic standing before they enrolled and their ultimate academic achievement. The study utilized 'O' level grades and data from the first three semesters[20]. Yıldız H et al.[21] utilized artificial neural networks to analyze students' final grades, midterm grades, and educational backgrounds. The objective was to forecast whether students would pass or fail. Asif R et al.[22] employed data mining methodologies to analyze student performance in two dimensions: firstly, to forecast student performance after four years of study, and secondly, to examine the progression of typical situations and integrate it with the projected outcomes to offer timely alerts to students. Macfadyen and Dawson[23] conducted a study using tracking data from Blackboard Vista to examine the influence of different online learning activities on learning outcomes. The study conducted by Zacharis et al.[24] focused on developing a predictive model for learning outcomes in blended courses using Moodle tracking data. Similarly, You et al.[25] examined the behaviors that have a significant impact on learning outcomes based on tracking data from a learning management system (LMS).

In summary, most of the existing related researches focus on static analysis of outcome data, with almost no dynamic research on process data. There are still some major issues that need further research related to the learning crisis in higher education:

1. Which components comprise the academic warning indicator system to ensure sufficiency?
2. What aspects constitute the comprehensiveness of the academic warning model?
3. Is the calculation method of academic warning sufficiently diverse?
4. Is the academic early warning technology accurate enough?

Taken together, the reviewed studies either rely on static feature sets or exhibit limited dynamic-update capacity, leaving a critical gap in timely academic-risk prediction. To address these shortcomings, a dynamic academic early warning system based on radial basis function (RBF) neural network is proposed, and the main contributions are concluded as follows:

1. The system firstly screened out the key factors affecting academic performance through questionnaires, and then the FCE-AHP method was adapted to sort out and construct the indicator system of academic early warning, and established the specific content of evaluation indicators.

2. Utilizing the ability of neural networks to approximate arbitrary continuous functions and their rapid convergence, the system is able to simulate and predict the nonlinear development of learning crises.
3. In addition, this study developed a neural network-based academic alert platform using HTML, CSS, JavaScript, Python, and PyTorch, aiming to provide customized alert services for students so as to identify and intervene in academic crises more effectively.
4. By combining four commonly used machine learning classification algorithms, a comprehensive academic early warning model is constructed, and a corresponding early warning system is developed. Through this study, it is expected to provide a new tool for universities to manage and support students' academic development in a more scientific and efficient way.

Our entire research process will be detailed in the sections below. Section "Introduction" focuses on introducing the topic's purpose. Section "Early warning indicator system construction" contains information about related research on the early warning indicator system. The academic influencing factors are mainly explained in this part. In Section "Construction of FCE", early warning model based on RBF is created, and in Section "Construct early warning model based on RBF", it is mainly provided the effectiveness of the constructed RBF neural network thorough comparison with other methods. Section "Model architecture flowchart" of the paper describes our approaches to solve the problem by our designed neural network-based academic alert platform. Finally, Section "Conclusions" contains the conclusions and future development directions we discovered during our research on this topic.

## Early warning indicator system construction

This study adopts fuzzy comprehensive evaluation (FCE) with analytic hierarchy process (AHP) method, which combines the fuzzy logic with hierarchical analysis to deal with the uncertainties and subjectivity of academic influencing factors.

### Academic influencing factors

In the preliminary stage of constructing the evaluation indicators system, specific information is clarified, and four majors in the college of engineering of a university are selected. A total of 1,245 students from freshmen to juniors are surveyed by questionnaires, with 1,231 valid questionnaires. A wide range of factors that may affect academic performance are meticulously screened and analyzed through the extensive questionnaires. Comprehensive consideration of students' learning behavior, learning environment, personal habits, psychological state and other factors are utilized to ensure the comprehensiveness and accuracy of the evaluation indicators. Only options with more than 15% selection are selected for this form survey. Thirty indicators are finalized, and the influencing factors selected are shown in Table 1 and Fig. 1.

### Adopt FCE-AHP method to construct the indicator system

FCE is an evaluation method based on fuzzy theory for dealing with the ambiguities and uncertainties of the evaluation object. FCE can comprehensively consider the impact of various factors such as learning attitude, learning method, personal situation and other factors on the academic performance of students, so as to more comprehensively assess the academic status of students. By quantifying the evaluation indicators and establishing the evaluation model, the FCE is used to reduce the subjectivity in the evaluation process and improve the objectivity of the evaluation. The FCE construction process is as follows:

## Construction of FCE

FCE usually includes the following steps:

1. Define evaluation indicators ($U$) and factor indicators ($V$) as follows: $U = \{u_1, u_2, ..., u_n\}$ and $u_g(g = 1, 2, ..., n)$ denotes the ith factor that affects the evaluation value of the indicators; $V = \{v_1, v_2, ..., v_m\}$ and $v_t(t = 1, 2, ..., m)$ denotes the jth rank indicates the evaluation value of the indicators. In this paper, 10 indicators closely related to the student's academic process are select as evaluation core elements to form the evaluation set $V = \{v_1, v_2, ..., v_{10}\}$ has gradually increasing evaluation scores.
2. Affiliation of individual evaluation indicators. CRITIC Method[26] is adopted to determine the weight of each evaluation indicator which reflect its importance in the academic process.
3. Establishment of fuzzy sets. The evaluation objects are divided into different fuzzy sets, and the affiliation function of each indicator is determined. The affiliation degree of the $n$th element in $U$ to the $m$th element in $V$, denoted by $\boldsymbol{R}$:

$$\boldsymbol{R} = \begin{bmatrix} r_{11} & r_{12} & \dots & r_{1m} \\ r_{21} & r_{22} & \dots & r_{2m} \\ \dots & \dots & \dots & \dots \\ r_{n1} & r_{n2} & \dots & r_{nm} \end{bmatrix}$$

Constructing a fuzzy comprehensive evaluation model. According to the characteristics of the specific problem, establish a fuzzy comprehensive evaluation model by $B$.

$$B = W_c \times \boldsymbol{R} = \{\omega_1, \omega_2, ..., \omega_n\} \times \begin{bmatrix} r_{11} & r_{12} & \dots & r_{1m} \\ r_{21} & r_{22} & \dots & r_{2m} \\ \dots & \dots & \dots & \dots \\ r_{n1} & r_{n2} & \dots & r_{nm} \end{bmatrix}$$

| Number | Indicator layers description |
|---|---|
| Indicator1 | Social network usage duration |
| Indicator2 | Number of credits for elective courses |
| Indicator3 | Online learning duration |
| Indicator4 | Family status |
| Indicator5 | Dormitory entry and exit time |
| Indicator6 | Violation and disciplinary action |
| Indicator7 | Language expression ability |
| Indicator8 | The frequency and duration of entering the library |
| Indicator9 | Academic homework grades |
| Indicator10 | Interpersonal relationship |
| Indicator11 | Timely submission of study assignments |
| Indicator12 | Exercise duration and frequency |
| Indicator13 | Psychological health assessment |
| Indicator14 | Frequency of participation in communication and discussion |
| Indicator15 | Social activity time |
| Indicator16 | Funding for learning materials |
| Indicator17 | Time for entering and exiting the cafeteria |
| Indicator18 | Subject interests and professional adaptability |
| Indicator19 | Extracurricular practice and internship experience |
| Indicator20 | Frequency of participating in extracurricular academic tutoring |
| Indicator21 | Learning environment |
| Indicator22 | Sleep adequacy |
| Indicator23 | Number of failed courses in the past |
| Indicator24 | Classroom attendance |
| Indicator25 | Participation in subject related competitions and activities |
| Indicator26 | Development and implementation of personal learning plans |
| Indicator27 | Participation in academic papers or research projects |
| Indicator28 | Weekly self-learning time |
| Indicator29 | Game duration |
| Indicator30 | Frequency of obtaining internship and practical opportunities |

**Table 1**. Academic warning indicator table.

$$\lambda = BV^T$$

where $W_c$ is the weights calculated according to CRITIC in this paper, $\boldsymbol{R}$ is the fuzzy relationship matrix, $B$ is the fuzzy affiliation degree, and $\lambda$ denotes the fuzzy evaluation estimation value, which indicates the goodness of the evaluation results.

## Construction and analysis of FCE

The weighting table of evaluation indicators was determined through the analysis of expert scoring questionnaire, as shown in Table 2:

From Table 2, it is shown that 10 indicators with the highest indicator weights are selected: class attendance (0.048), development and implementation of personal study plans (0.048), participation in discipline-related competitions and activities (0.046), learning environment (0.046), frequency of internships and practicums acquired (0.043), past failing number of courses (0.042), participation in academic papers or research projects (0.042), weekly independent study time (0.042), sleep adequacy (0.041), and length of time playing games (0.04).

## Construction of AHP

The AHP method can more accurately identify and quantify the degree of influence of various factors on academic performance and provide effective academic early warning based on the results of the expert evaluations and the calculation of weights. Through the AHP method, subjectivity can be reduced and the objectivity and accuracy of evaluation can be improved.

The 10 indicators selected through the principle of maximum FCE weights were assigned to the AHP method as shown in 2.2.2. A number of relevant professional teachers and experts were also invited to assign importance to the guideline and program layers and construct judgment matrices. Different indicators have different influence on experts' scores, so the weight of each indicator is calculated and then evaluated comprehensively to make the final evaluation results closer to the facts. The steps are shown below.

(1) Constructing a judgment matrix $\boldsymbol{A}$

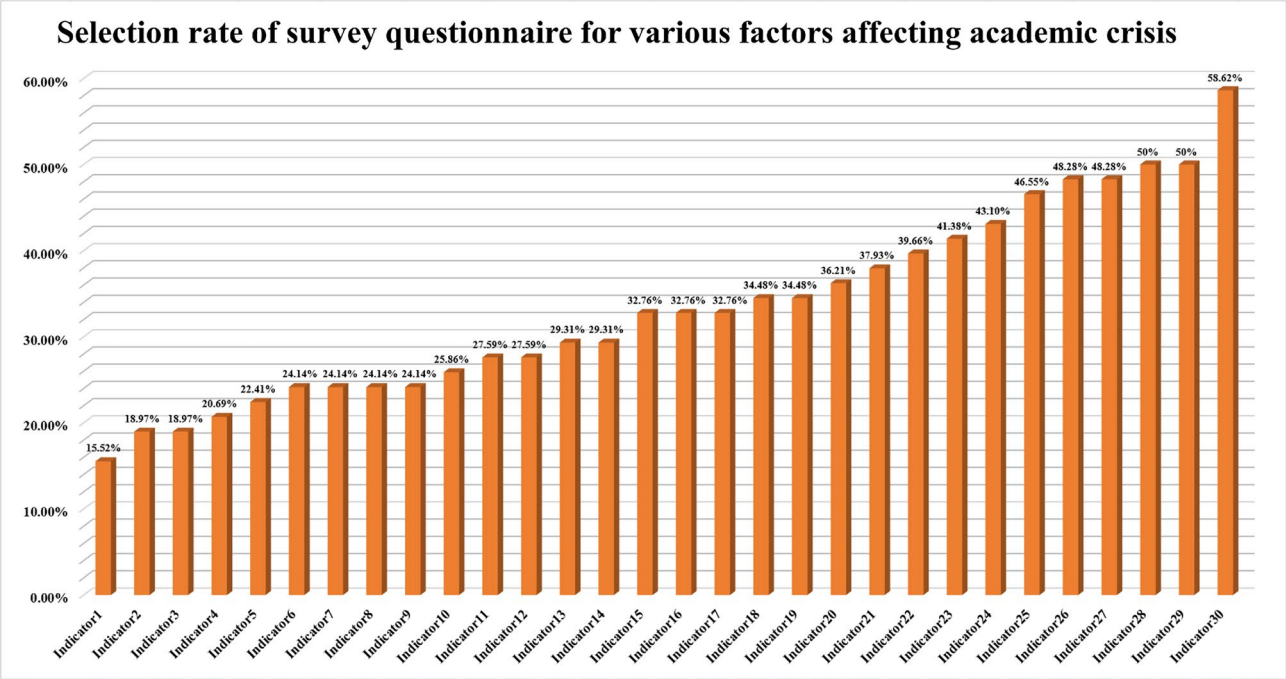



**Fig. 1**. Selection rate of survey questionnaire for various factors affecting academic crisis.

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{bmatrix}$$

where, $a_{ij}$ denotes an element of the judgment matrix that represents the relative importance of factor $i$ relative to factor $j$ in the criterion or program level, and the value of $a_{ij}$ is usually determined through expert scoring, collective discussion, or based on empirical judgment. The value of $a_{ij}$ is usually determined through expert scoring, collective discussion or based on empirical judgment. A relative importance indicator rating scale can be used to assist in assigning values. Relative Importance Indicator Judgment Table is shown in Table 3.

(2) Vector normalization.

First, normalization was performed according to the formula:

$$\vec{a}_{ij} = a_{ij} \Big/ \sum\nolimits_{i=1}^{n} a_{ij}$$

where,$i, j = 1, 2, 3, \ldots, n$.

Second, summing the elements of the matrix yields

$$\vec{W}_i = \sum\nolimits_{i=1}^{n} \quad a_{ij} \quad (i, j = 1, 2, \ldots, n)$$

Third, $\vec{W}_i$ is then normalized to

$$W_i = \quad \vec{W}_i \Big/ \sum\nolimits_{i=1}^{n} \vec{W}_i \quad (i = \ 1, 2, \ldots, n)$$

(3) Consistency test.

The consistency test is performed on the judgment matrix by defining the consistency indicator $CI = \lambda_{\max}/(n-1)$, $\lambda_{\max}$ as the largest characteristic root of the judgment matrix, and $\lambda_{\max} = \sum\limits_{i=1}^{n} [A\vec{W}] \Big/ n\omega_i$

. As to the consistency ratio $CR = CI/\,RI$, the value of the random consistency indicator $RI$ is corresponding to the order *n* of the matrix. The judgment matrix is considered to be consistent if CR < 0.1.

Relying on scholars and relevant experts to assign importance to the 10 indicators in the criterion layer, the criterion layer judgment matrix $\boldsymbol{A}$ is constructed, and in order to improve the readability of the table, its importance assignment is shown in Tables 4and 5.

Based on the judgment matrix $\boldsymbol{A}$, the results of AHP were obtained and analyzed as shown in Table 6.

| Indicator layer | Weight |
|---|---|
| Psychological health assessment | 0.027 |
| Development and implementation of personal learning plans | 0.048 |
| Classroom attendance | 0.048 |
| Sleep adequacy | 0.041 |
| Weekly self-learning time | 0.042 |
| Social activity time | 0.025 |
| Subject interests and professional adaptability | 0.027 |
| Interpersonal relationship | 0.024 |
| Social network usage duration | 0.029 |
| Frequency of participation in communication and discussion | 0.026 |
| Dormitory entry and exit time | 0.026 |
| Family status | 0.032 |
| Participation in subject related competitions and activities | 0.046 |
| Participation in academic papers or research projects | 0.042 |
| Frequency of obtaining internship and practical opportunities | 0.043 |
| Number of credits for elective courses | 0.016 |
| Online learning duration | 0.014 |
| Frequency of participating in extracurricular academic tutoring | 0.035 |
| Learning environment | 0.046 |
| Violation and disciplinary action | 0.023 |
| Language expression ability | 0.032 |
| Extracurricular practice and internship experience | 0.025 |
| Funding for learning materials | 0.035 |
| Timely submission of study assignments | 0.039 |
| Number of failed courses in the past | 0.042 |
| Time for entering and exiting the cafeteria | 0.028 |
| Game duration | 0.04 |
| The frequency and duration of entering the library | 0.033 |
| Exercise duration and frequency | 0.031 |
| Academic homework grades | 0.037 |

**Table 2**. Evaluation indicator weight table by expert scoring questionnaire.

| Scale | Meaning |
|---|---|
| 1 | Two factors are equally important compared to each other |
| 3 | Compared to two factors, the former factor is significantly more important than the latter factor |
| 5 | Compared to two factors, the former factor is more important than the latter factor |
| 7 | Compared to two factors, the former factor is extremely important compared to the latter factor |
| 9 | Compared to two factors, the former factor is extremely important compared to the latter factor |
| 2, 4, 6, 8 | Middle value of two adjacent scales |
| The reciprocal of the above values | Comparing the two factors in reverse is the reciprocal of the original comparison |

**Table 3**. Relative importance ratio table.

In addition, this judgment matrix gets maximum eigenvalue $\lambda_{max}$, and consistency test is performed:

$$CI = \frac{\lambda_{\max} - n}{n-1} = \frac{11.24 - 10}{10 - 1} = 0.138$$

Consistency metrics $RI = 1.486$, the randomized consistency ratio $CR$ is shown below:

$$\mathrm{CR} = \frac{CI}{RI} = \frac{0.138}{1.486} = 0.093 < 0.10$$

The $CR$ value is 0.093 (<0.1), which passes the consistency test (see Table 7), indicating a more reasonable judgment matrix for AHP. The $RI$ values for matrices of different orders are shown in Table 8.

| Number | Indicator description |
|---|---|
| IN1 | Classroom attendance |
| IN2 | Development and implementation of personal learning plans |
| IN3 | Participation in subject related competitions and activities |
| IN4 | learning environment |
| IN5 | Frequency of obtaining internship and practical opportunities |
| IN6 | Number of failed courses in the past |
| IN7 | Participation in academic papers or research projects |
| IN8 | Weekly self-learning time |
| IN9 | Sleep adequacy |
| IN10 | Game duration |

**Table 4**. Serial numbers of various indicators.

| Indicator | IN1 | IN2 | IN3 | IN4 | IN5 | IN6 | IN7 | IN8 | IN9 | IN10 |
|---|---|---|---|---|---|---|---|---|---|---|
| IN1 | 1 | 0.5 | 1 | 0.588 | 1.429 | 1.25 | 0.769 | 1 | 1 | 1 |
| IN2 | 2 | 1 | 2 | 2 | 1 | 0.5 | 0.588 | 0.5 | 2 | 0.833 |
| IN3 | 1 | 0.5 | 1 | 0.5 | 1 | 2 | 2 | 1 | 1 | 0.333 |
| IN4 | 1.7 | 0.5 | 2 | 1 | 0.714 | 1 | 2 | 0.667 | 1 | 5 |
| IN5 | 0.7 | 1 | 1 | 1.4 | 1 | 1 | 0.769 | 2 | 1.111 | 2 |
| IN6 | 0.8 | 2 | 0.5 | 1 | 1 | 1 | 1 | 1 | 0.5 | 2 |
| IN7 | 1.3 | 1.7 | 0.5 | 0.5 | 1.3 | 1 | 1 | 1 | 2 | 1.111 |
| IN8 | 1 | 2 | 1 | 1.5 | 0.5 | 1 | 1 | 1 | 0.5 | 1 |
| IN9 | 1 | 0.5 | 1 | 1 | 0.9 | 2 | 0.5 | 2 | 1 | 1 |

**Table 5**. Importance assignment.

| Influence factor | Weight value(%) | Maximum eigenvalue | CI value |
|---|---|---|---|
| IN1 | 8.526 | 11.24 | 0.138 |
| IN2 | 11.126 | | |
| IN3 | 9.375 | | |
| IN4 | 12.831 | | |
| IN5 | 10.531 | | |
| IN6 | 9.511 | | |
| IN7 | 10.244 | | |
| IN8 | 9.396 | | |
| IN9 | 9.691 | | |
| IN10 | 8.769 | | |

**Table 6**. AHP analytic hierarchy process results.

| Maximum eigenvalue | CI value | RI value | CR value | Consistency test results |
|---|---|---|---|---|
| 11.24 | 0.138 | 1.486 | 0.093 | Pass |

**Table 7**. Consistency check.

In addition, it also passed the consistency test, yielding a consistency ratio CR value of 0.093 (< 0.1), indicating that the constructed AHP judgment matrix has good consistency and high confidence.

## Analysis of results using the FCE-AHP approach

The flowchart of the FCE-AHP method is shown in Fig. 2. By combining with FCE-AHP to screen and assign weights to the factors influencing the academic crisis, this study finalized 10 key factors from 30 initial indicators to be used as input data and play a decisive role in the prediction accuracy of the academic early warning model. The weights assigned to the 10 key influencing parameters are shown in Table 6.

| n | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| RI | 0 | 0 | 0.56 | 0.90 | 1.12 | 1.24 | 1.32 | 1.41 | 1.45 | 1.49 | 1.52 |

**Table 8**. RI values corresponding to matrix order ***n.***

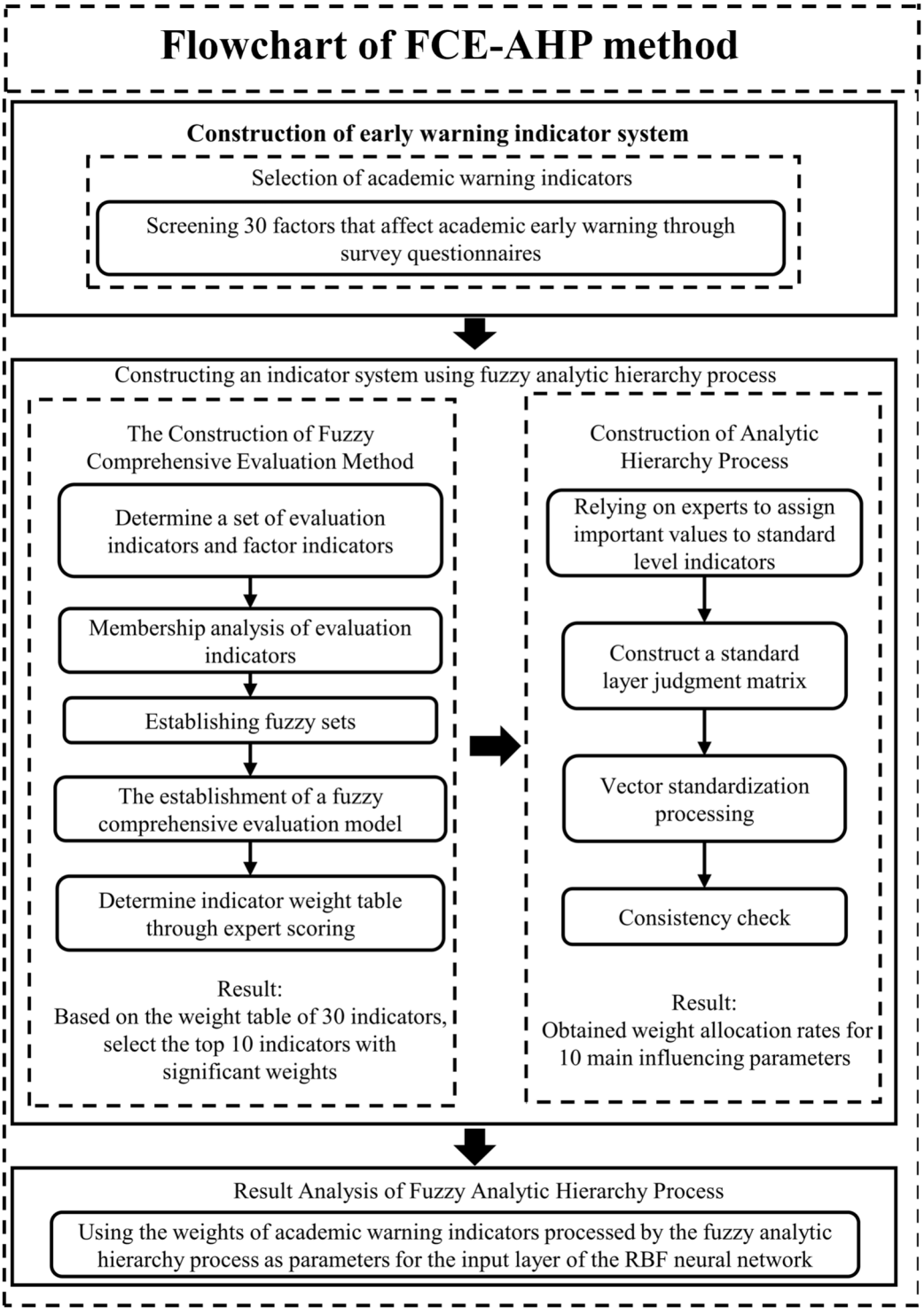



**Fig. 2**. Flowchart of FCE-AHP method.

Through the FCE-AHP method, the researcher is able to analyze the degree of influence of each factor on academic performance and construct a scientific and reasonable academic warning indicator system. The method first decomposes the academic influencing factors into different levels by constructing a hierarchical structural model, and then compares and quantifies the factors using expert scores and fuzzy logic to determine their relative importance.

In terms of the primary outcome, the application of the FCE-AHP method enabled the researcher to filter out the main factors that have the greatest impact on academic performance, such as class attendance, strength of implementation of individual study plans, and participation in subject competitions. These factors were given high weights and used as key input parameters for the academic early warning model.

## Construct early warning model based on RBF

In the academic crisis analysis, the main parameters of the academic warning model have been determined. Considering the large amount of academic warning data, nonlinear changes and the characteristics of warning categorization, it was decided to use RBF neural network to achieve accurate warning of students in the learning process.

RBF neural network is a powerful nonlinear modeling tool that can effectively capture complex relationships in data and has good generalization ability. By using an appropriate number and location of radial basis functions in the network, it is possible to effectively partition the data in a high-dimensional space, thus enabling accurate prediction and early warning of students' academic performance.

The basic structure of RBF neural network consists of three parts: input layer, hidden layer and output layer. The input layer is used to receive the student sample data, i.e. evaluation indicator parameters. The hidden layer calculates the response signal according to the radial basis function, and the output layer synthesizes the response signal and outputs the final result, whose topology is shown in Fig. 3:

### Data pre-processing

Based on the data related to students in the Oracle database obtained from the information center of a school, the required data were extracted by concatenated table query through SQL statements and imported into the academic alert database, which provided a solid data foundation for the research and system development of the academic alert model.

When faced with structured data from different data tables alone, the data was found to have problems such as incompleteness, missing samples and duplicates, and thus had to be pre-processed.

Data pre-processing includes steps such as data cleaning, integration, transformation, normalization, and attribute creation. Using the Pandas library in Python, the survey data was read in and divided into training and test sets at a ratio of 4:1. Subsequently, normalization was performed on the dataset to eliminate the effect of magnitude and ensure the generalization ability of the model. The normalization formula is as follows:

$$x_{\text{norm}} = \frac{x_{\text{in}} - \min(x_{\text{in}})}{\max(x_{\text{in}}) - \min(x_{\text{in}})}$$

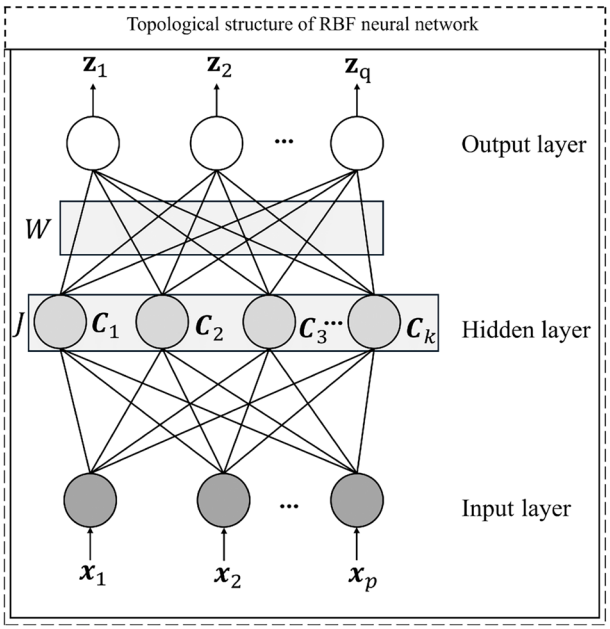



**Fig. 3**. Topological structure of RBF neural network.

$x_{\text{in}}$ represents value of a feature in original dataset. $x_{\text{norm}}$ is the data after normalization. $\min(x_{\text{in}})$ is the minimum value and $\max(x_{\text{in}})$ is the maximum value.

## Model training

The model training is mainly divided into five parts: input layer computation, hidden layer computation, output layer computation, comparative analysis of the loss function descent process under different activation functions, and model parameter optimization. The specific construction process is as follows:

*Input layer calculation*

The input layer of the model directly inputs student sample data through a $X_l \times W_l$ matrix that receives student sample data directly. $c_j$ is the dimensions of input vector. $X_l = (x_1, x_2, ..., x_l)$, i.e., and it is also equal to the number of neurons in the input layer. In this context, $l = 10$. The $W_l$ are the weights of the samples subjected to the AHP hierarchical analysis method, which is the connection between the data and the input layer. The specific values of these weights are presented in Table 6.

*Hidden layer calculation*

In RBF neural network, the nodes in the hidden layer use the radial basis function as an activation function to respond to the input data. When a Gaussian function is chosen as the radial basis function, the hidden layer nodes $j$ responds to the input vector $X_l$. The output of the hidden layer node $h_j(X_l)$ can be expressed as:

$$\partial\varphi(X_l, c_j) = h_j = \exp\left(-\frac{\|X_l - c_j\|^2}{2\sigma_j^2}\right)$$

In (11), $c_j$ is the center vector of the hidden layer node. $\|X_l - c_j\|$ denotes the Euclidean distance between the input vector $X_l$ and center vector $c_j$. $\sigma_j$ is the width parameter of the radial basis function, also known as the scale parameter or scale factor. The NEWRB function is utilized in MATLAB to design an approximate RBF network, setting the network target mean square error as $10^{-8}$. According to the results, the number of nodes in the hidden layer is taken as $N_h = 8$.

*Output layer calculation*

The warning levels of the academic crisis are normal, concern, serious and dangerous. The output range of RBF neural network is 1, 2, 3 and 4, which corresponds to the four warning levels by rounding, so the number of nodes in the output layer is 1. The output value of this node is.

$$y = W_k * h_j + b_{2k}$$

where. $W_k$ is the initialized weights, and $h_j$ is the output of the hidden layer, and $b_{2k}$ is the initialized bias. $y$ denotes the output value of the unpassed activation function of the output layer.

Since this is a multi-classification problem, a detailed comparison between three output layer activation functions, **Tanh**, **Sigmoid** and **Relu**, is performed. The detailed process of loss value variation is shown in Fig. 4. It is observed that the **Relu** activation function may show high loss values at the beginning of training, which may be due to its zero output for negative inputs, a phenomenon commonly referred to as "neuron death". The **Tanh** activation function, on the other hand, may exhibit large fluctuations in the early stages of training, which may be due to its high sensitivity to the input data. On the contrary, the **Sigmoid** activation function has a smoother change in loss values throughout the training process, which may be due to its output range between 0 and 1, which has better properties for gradient propagation and parameter updating. Therefore, based on

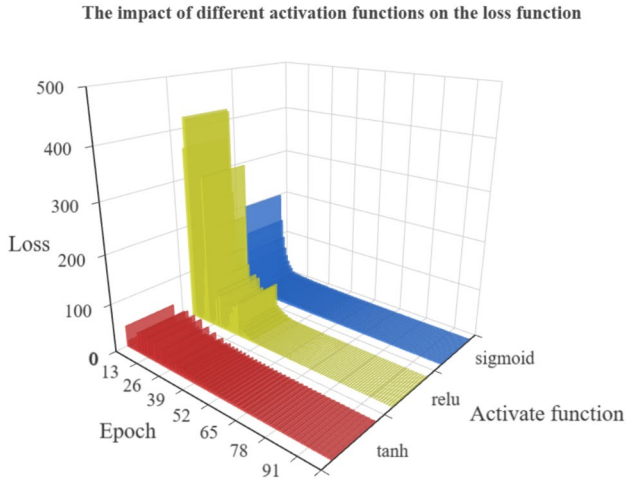



**Fig. 4**. Descending process of loss function under different activation functions.

the observation, the **Sigmoid** activation function seems to be more suitable for the task requirements and the Sigmoid activation function is used to output the final result

$$\sigma(y) = 1/\left[1 + \exp(-y)\right]$$

The four classifications of academic crisis can be obtained, thus providing a decision-making basis for learning warning intervention. Randomly initialize the weights $W_k$ and bias $b_{2k}$.

## Optimization of model parameters

Using stochastic gradient descent (SGD) to optimize model parameters. The objective function is calculated using the following formula:

$$E(W_k, b_{2k}; y, y^*) = \left\| h_{W_k, b_{2k}}(y) - y^* \right\|^2 \Big/ 2$$

$$h_{W_k, b_{2k}}(y) = \sigma(y) = 1/\left[1 + \exp(-y)\right] = 1/[1 + \exp(W_k * h_j + b_{2k})]$$

where $y^*$ is the reference value of the sample.

Calculate the gradient of output layer parameters using the formula:

$$\delta^L = \partial E \Big/ \partial u^L = \partial E / \partial h_{W_k, b_{2k}}(y) = h_{W_k, b_{2k}}(y) - y^*$$

$$\partial E \Big/ \partial W_k^L = \partial E \Big/ \partial u^L \bullet \partial u^L \Big/ \partial W_k^L = \delta^L \bullet h^{L-1}$$

$$\partial E \Big/ \partial b_{2k}^L = \partial E \Big/ \partial u^L \bullet \partial u^L \Big/ \partial b_{2k}^L = \delta^L$$

where $\delta^L$, $L$ represents the gradient of layer $L$, $E$ represents the objective function, $u^L$ represents the output of layer $L$, $x$ represents the input of layer $L$, $y^*$ represents the true value, $W^L$ represents the weight parameter of layer $L$, $b^L$ represents the bias parameter of layer $L$, and $h^{L-1}$ represents the output of layer $^{L-1}$.

Calculate the gradient of hidden layer parameters using the formula:

$$\begin{aligned}
\Delta c_j &= \partial E / \partial c_j = \partial E / \partial \varphi(X_l, c_j) \bullet \partial \varphi(X_l, c_j) / \partial c_j \\
&= 1/m \bullet \sum_{i=1}^{m} \left(h_{W_k, b_{2k}}(y) - y^*\right) \bullet W \bullet \partial \varphi(X_l, c_j) / \partial c_j \\
&= 1/m \bullet \sum_{i=1}^{m} \left(h_{W_k, b_{2k}}(y) - y^*\right) \bullet W \varphi(X_l, c_j) \bullet (X_l - c_j) / \sigma_j^2 \\
&= 1/m\sigma_j^2 \bullet \sum_{i=1}^{m} \left(h_{W_k, b_{2k}}(y) - y^*\right) \bullet W \bullet \varphi(X_l, c_j) \bullet (X_l - c_j)
\end{aligned}$$

$$\begin{aligned}
\Delta \sigma_j &= \partial E / \partial \sigma_j = \partial E / \partial \varphi(X_l, c_j) \bullet \partial \varphi(X_l, c_j) / \partial \sigma_j \\
&= 1/m \bullet \sum_{i=1}^{m} \left(h_{W_k, b_{2k}}(y) - y^*\right) \bullet W \bullet \partial \varphi(X_l, c_j) / \partial \sigma_j \\
&= 1/m\sigma_j^3 \bullet \sum_{i=1}^{m} \left(h_{W_k, b_{2k}}(y) - y^*\right) \bullet W \bullet \partial \varphi(X_l, c_j) \bullet \left\| (X_l - c_j) \right\|^2
\end{aligned}$$

where $\Delta c_j$ is the gradient of the center point of the hidden layer neuron, and $c_j$ is the center point of the $j$ neuron, $\partial \varphi(X_l, c_j)$ is a Gaussian Sum function, and $m$ is the number of samples, $\sigma_j$ represents the Gaussian kernel width of the $j$ neuron.

The final formula for updating parameters is:

$$\sigma_{k+1} = \sigma_k - \eta \Delta \sigma$$

$$c_{k+1} = c_k - \eta \Delta c$$

where $\sigma_{k+1}$ is the updated Gaussian kernel width, $\sigma_k$ is the Gaussian kernel width before the update, $\eta$ is the learning rate that determines the step size of parameter updates, $\Delta\sigma$ It is the gradient of the Gaussian kernel width. $c_{k+1}$ is the updated neuron center point, and $c_k$ is the neuron center point before the update, $\Delta c$ is the gradient of the neuron center point.

Since it is a minimization objective function, it is based on each parameter $\theta$. Use the negative direction of the gradient to update each $\theta$:

$$\partial / \partial \theta_j \bullet J(\theta) = \partial / \partial \theta_j \bullet 1/2 \bullet \left(h_\theta(x) - y\right)^2 = \left(h_\theta(x) - y\right) x_j$$

where $\partial / \partial \theta_j$ represents taking partial derivatives for each component, $h_\theta(x)$ Represents the forward propagation process of a neural network, where $x_j$ represents the feature in column $j$.

In each iteration, according to the parameter $\theta$ To the loss function $j(\theta; x_i; y_i)$ at a single data point $(x_i, y_i)$. The first-order derivatives are computed on a single data point, and then the learning rate is calculated in terms of the $\eta$ . The size of the parameter $\theta$ is updated. This process can be represented by the following equation:

$$\theta = \theta - \eta \nabla_\theta j(\theta; x_j; y_j)$$

In (22), $\eta$ denotes the learning rate, which determines the step size of the parameter update, and $\nabla_\theta j(\theta; x_i; y_i)$ denotes the loss function on each piece of data for the parameter $\theta$ calculated gradient. By updating the parameters $\theta$ , the model can be gradually optimized so that the loss function reaches the minimum value, thus improving the performance and generalization ability of the model.

The process of updating the weights of RBF neural network based on SGD is shown in Fig. 5:

## Model architecture flowchart

The flowchart of the academic early warning system was successfully constructed through the analysis, and this flowchart demonstrates the key steps and operational logic of the early warning model, which is shown in Fig. 6:

## Validation of effectiveness of the construction RBF neural network

*Functional realization*

Based on the academic early warning model, this study conducted a comprehensive assessment of students' academic status. This assessment process is mainly divided into two key stages: model parameter setting and model testing, and the specific prediction process is detailed in Fig. 7.

*Model parameterization*

According to the requirements of the early warning model, parameters including the number of input data, the number of nodes in the implicit layer, the number of nodes in the input layer, the maximum number of training times, the stopping training condition and the learning rate are set to optimize the model performance. The specific settings are shown in Table 9 below:

*Model training*

For this training, 1000 out of 1100 students in the Department of Electronics and Information Technology of the school's 2018 and 2019 cohorts were selected to train the model with their personal behavioral data and graduation during their three years of schooling. Among them:

1. School one data totaled 625,896,124 entries, collected for one main indicator: Class Attendance.
2. A total of 886,034,856 pieces of web data from the school's academic affairs platform are collected for three main indicators: Discipline-related competition and activity participation, frequency of internship and practicum opportunity acquisition, and academic paper or research project participation.
3. Historical student achievement data totaled 783,648 entries, collected for one main indicator: Number of courses failed in the past.
4. The total number of questionnaires was 1,284, and five main indicators were collected: Learning environment, development and implementation of personal study plan, weekly independent study time, sleep adequacy, and hours of game playing.

After data preprocessing, 11 sets of personal behavioral characteristics data for 1100 students were obtained, which constituted the raw dataset for academic prediction. The 1100 students in the original dataset were split

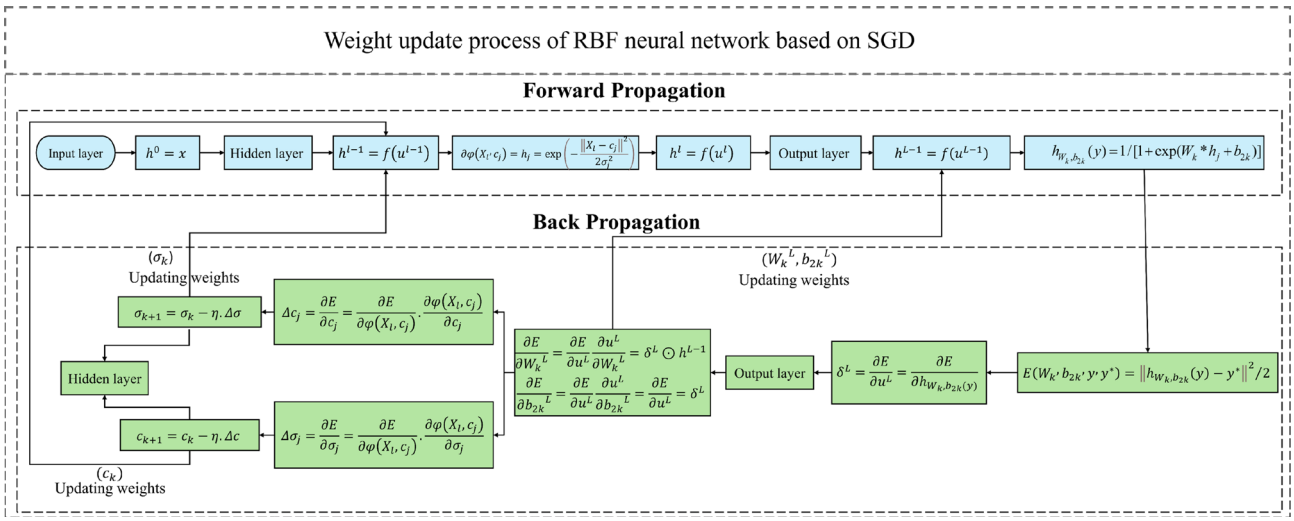



**Fig. 5**. Weight update process of RBF neural network based on SGD.

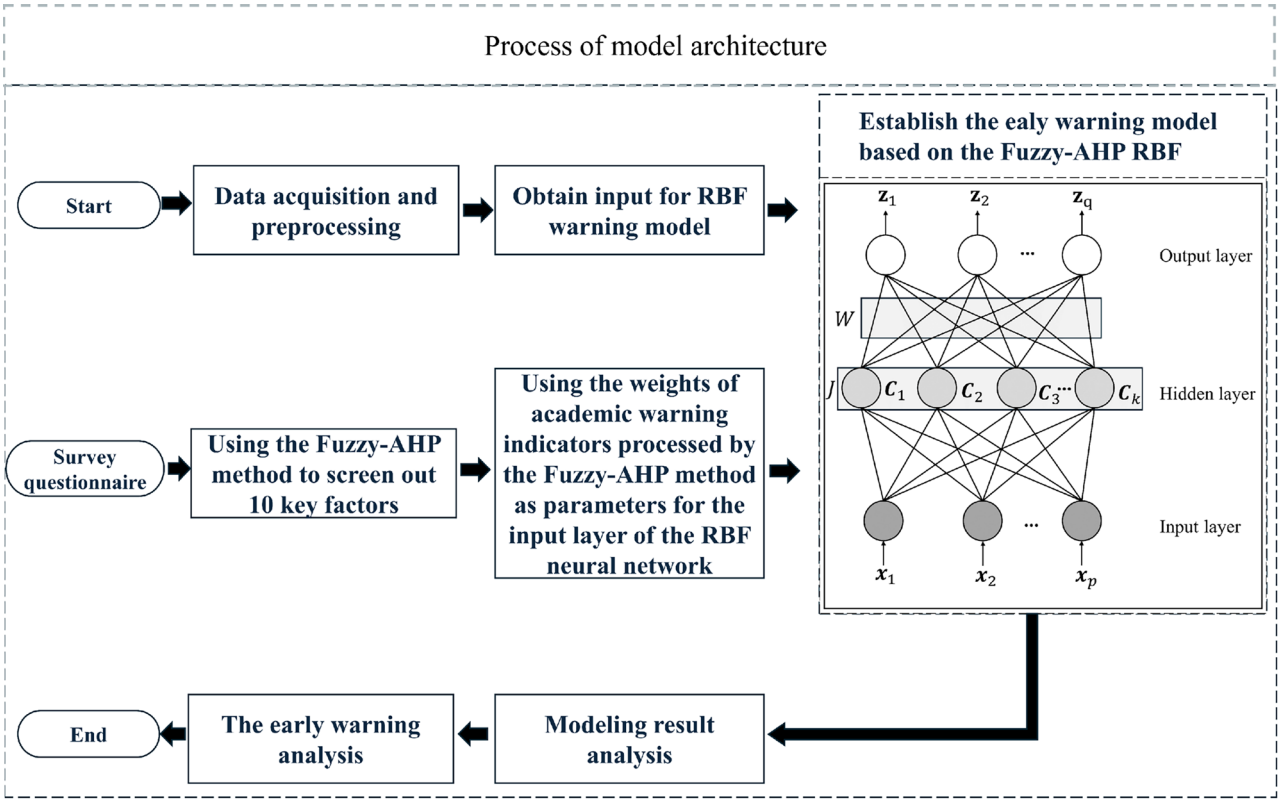



**Fig. 6**. Process of model architecture.

into a training dataset and a measurement set, where the dataset in the training dataset contained 1000 students. After training on this training dataset, the training loss function obtained is shown in Fig. 8. In Fig. 8a, $W_1$ represents weight of the hidden layer, and $W_2$ represents weight of the output layer.

During the training process, the loss function exhibits a gradual decrease with the increase in the number of iterations and eventually converges to a low and stable value. Specifically, after about 100 cycles of training, the decreasing rate of the loss function significantly slows down and flattens out, which indicates that the weights and biases of the model have been effectively optimized by stochastic gradient descent to a local minimum, i.e., the model achieves a good fit on the training set.

## Model testing

In this training, 100 out of 1100 students from the Department of Electronics and Information Technology of the school's 2018 and 2019 cohorts were selected to test the model with their personal behavioral data and graduation in the three years of schooling. The results of the test are shown in Fig. 9.

It can be seen that the predicted values are very close to the actual values, and only 9 samples have a significant errors between the predicted values and the actual values, indicating that the model has a good generalization ability, and the prediction results are more desirable to be applied to crisis intervention.

In order to demonstrate the effectiveness of the proposed RBF method, an in-depth comparative analysis of the performance of RBF neural networks, Recurrent Neural Networks (RNN), and Softmax regression in the task of academic alerting is conducted. The RNN network structure[27] is shown in Fig. 10. The Softmax regression network structure[28] is shown in Fig. 11.

A comprehensive evaluation is conducted based on key indicators such as $F1$ metric, accuracy and mean squared[29]. The $F1$ metric, as the harmonic mean of *Precision* and *Recall*, is utilized for assessing the model's performance in binary classification tasks. Accuracy is the proportion of correctly predicted samples to the total number of samples in the field of computer science. Mean Squared Error is a commonly used performance metric in regression problems to quantify the difference between predicted and actual values.

(1) $F1$ metric is a binary classification model for evaluating performance, which is the harmonic mean of *Precision* and *Recall*. *Precision* is the ratio of the number of samples correctly predicted by the model to be positive to the total number of samples predicted by the model to be positive. *Recall* is the ratio of the number of samples correctly predicted by the model to be positive to the total number of samples that are actually positive.

The formula for the $F1$ metric is as follows:

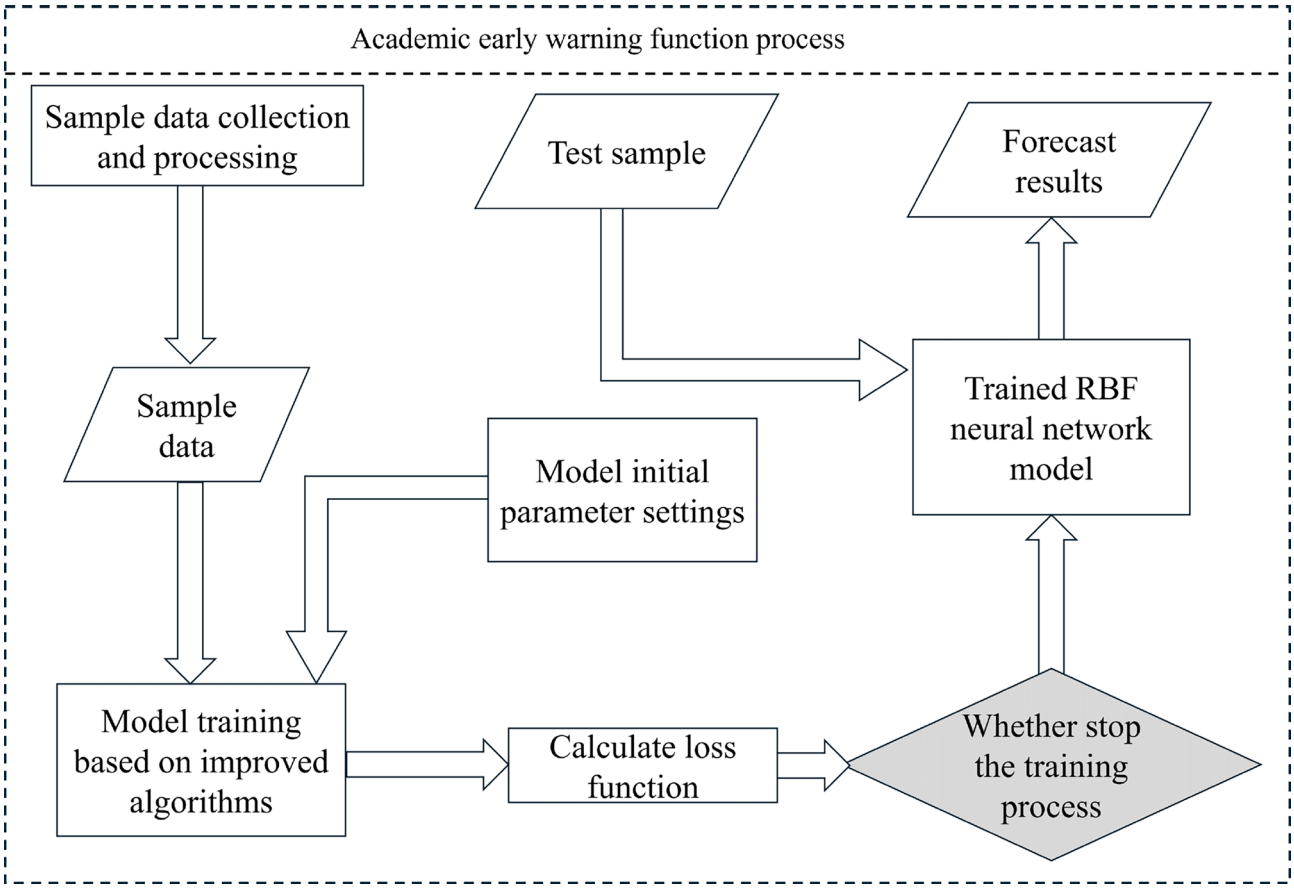



**Fig. 7**. Academic early warning function process.

| Parameter | Numerical value |
|---|---|
| Input data quantity | 1000 |
| Number of hidden layer nodes | 8 |
| Number of input layer nodes | 10 |
| Number of output layer nodes | 1 |
| Maximum training times | 100 |
| Stop training conditions | 0.0065 |
| Learning rate | 0.001 |

**Table 9**. System parameter setting.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

where the formulas for *Precision* and *Recall* are respectively:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

where $TP$ represents the number of true examples, and $FN$ represents the number of false negative examples, and $FP$ represents the number of false positive examples.

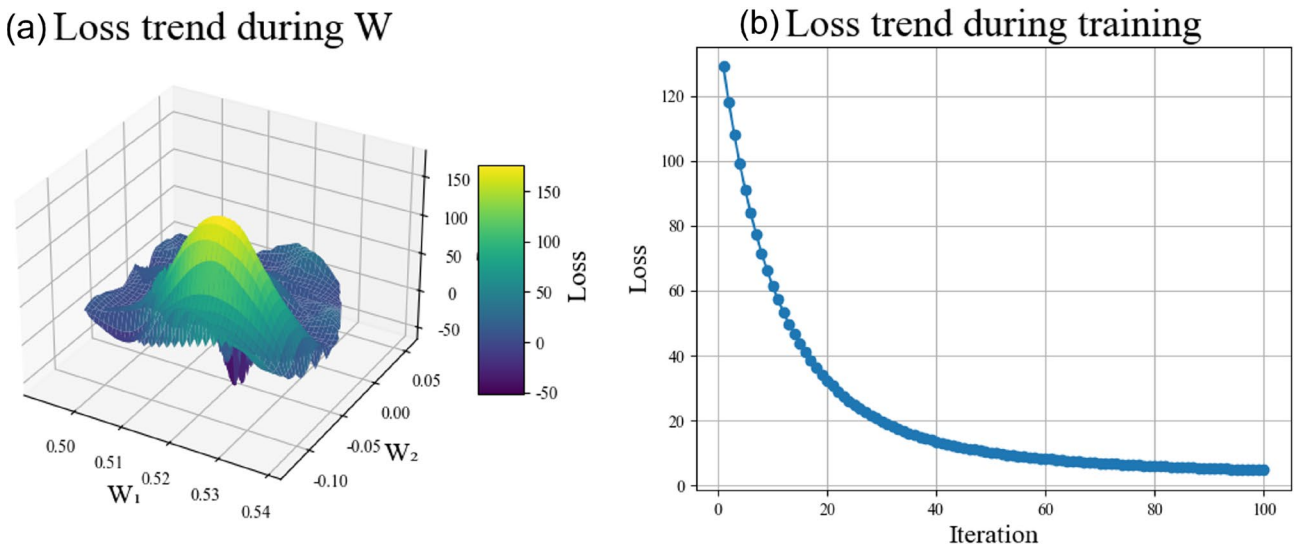



**Fig. 8**. (**a**) Loss trend during W; (**b**) Loss trend during training.

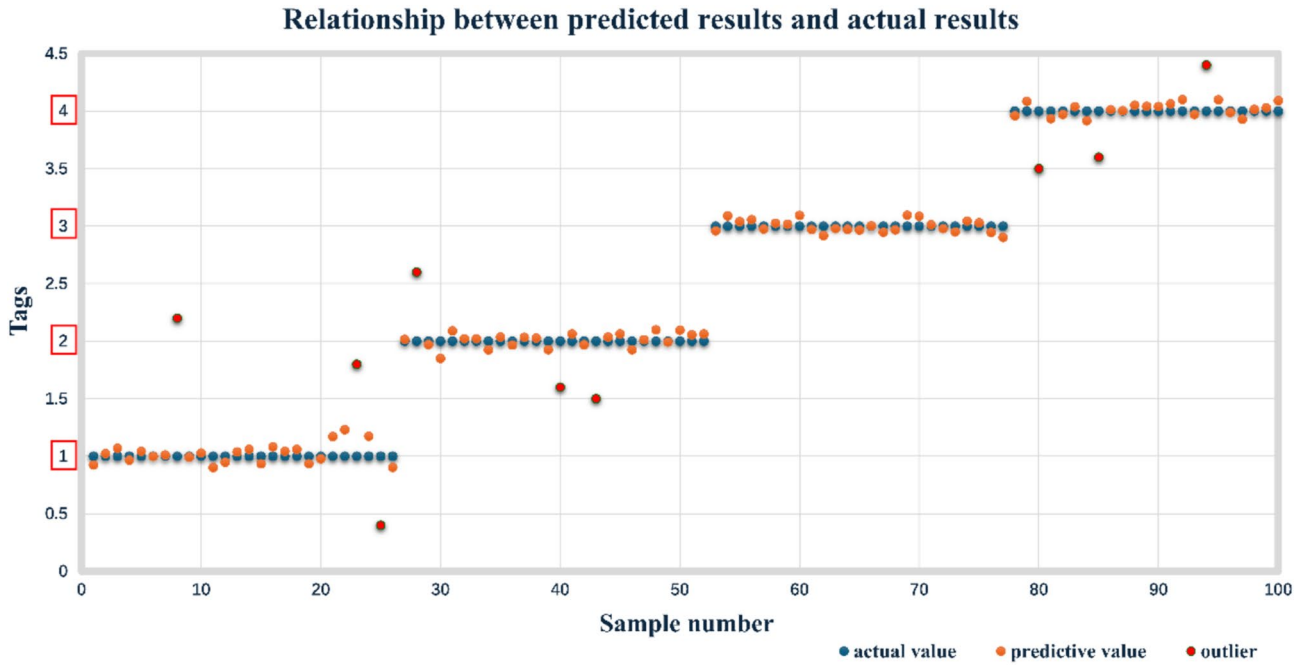



**Fig. 9**. Test sample results.

(2) *Accuracy* is one of the most intuitive performance metrics, indicating the proportion of all predictions made by the mode. It is calculated by the formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where $TN$ represents the number of true negative examples. The *Accuracy* is straightforward but may not be accurate enough in the case of category imbalance because it does not distinguish different types of errors.

(3) Mean square error (MSE))

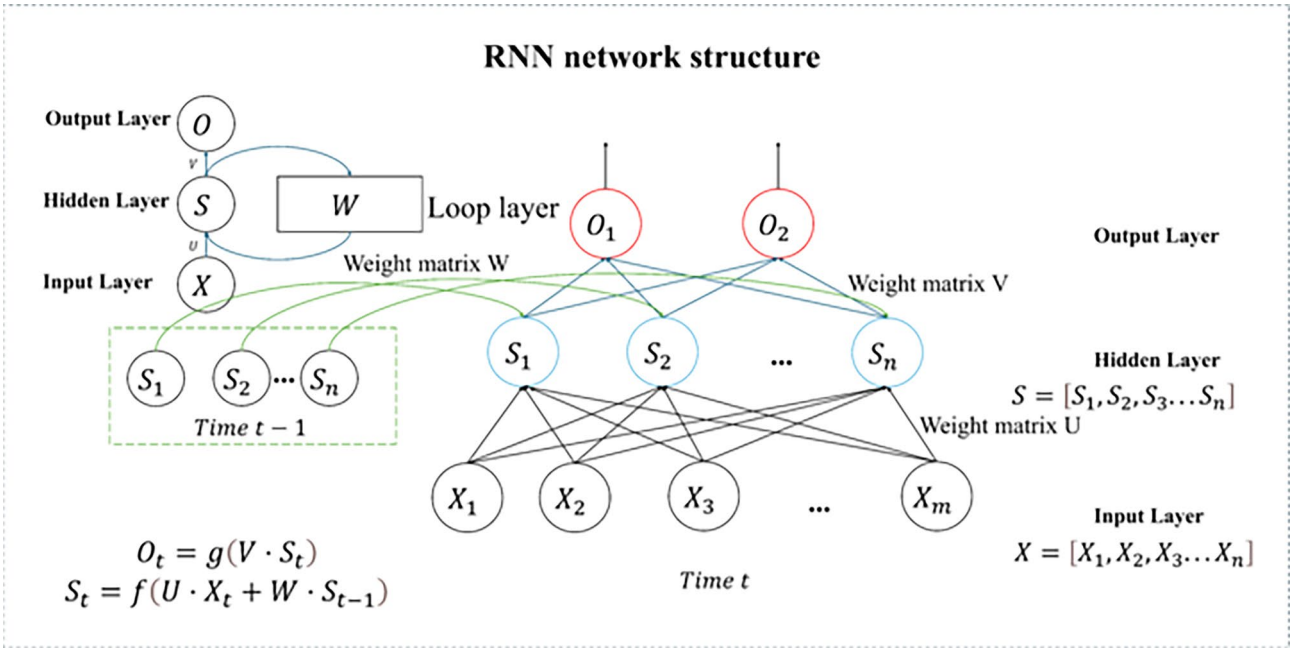



**Fig. 10**. RNN network structure.

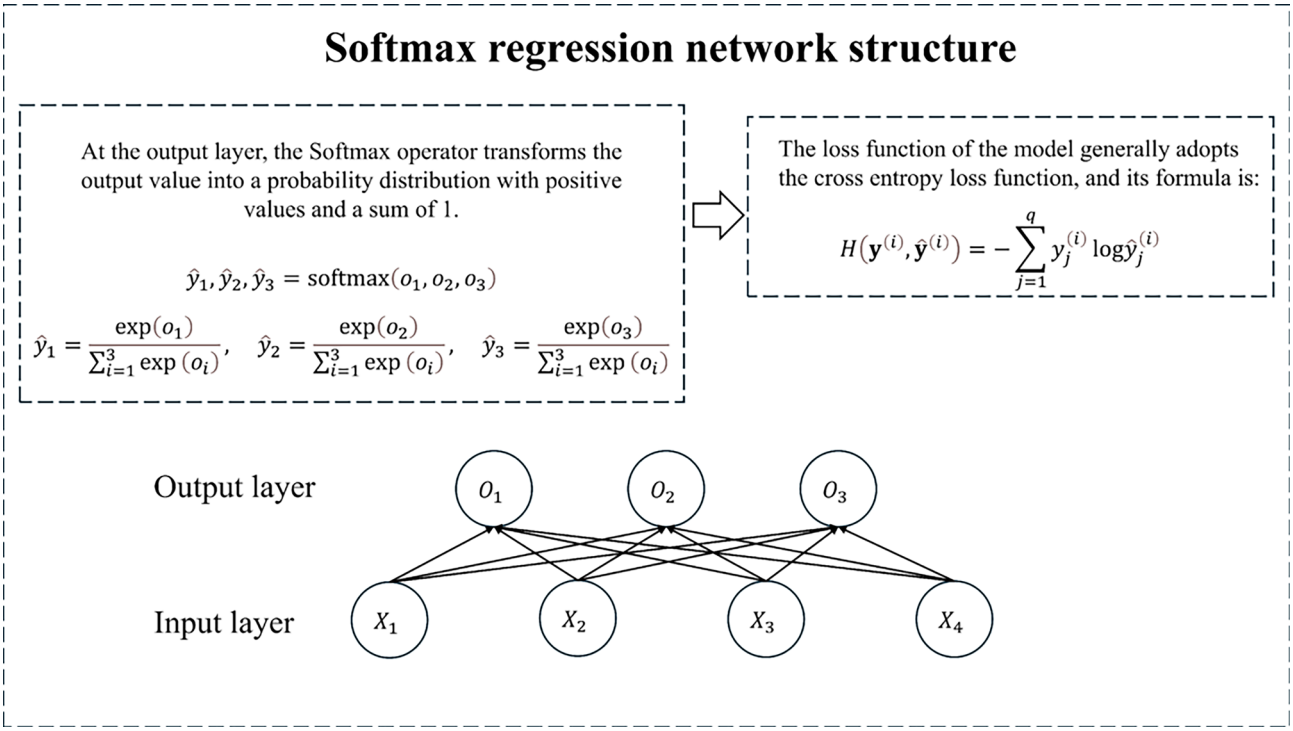



**Fig. 11**. Softmax regression network structure.

The mean square error is one of the most commonly used performance indicators in regression modeling, which is used to measure the difference between the predicted value and the actual value of the model, and its formula is as follows:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \widehat{Y}_i)^2$$

where $Y_i$ is the actual value of the first sample, $\widehat{Y_i}$ is the value of the $i$-th sample predicted by the model, and $n$ is the total number of samples. MSE measures the accuracy of prediction by calculating the average of the sum of squares of the prediction errors, and gives more weight to the larger errors.

(4) Root mean square error (RMSE)

RMSE is one of the most frequently adopted performance indicators in regression modelling. It is defined as the square-root of the Mean Square Error and is given by

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

where $y_i$ is the true value of the $i$-th sample, $\hat{y}_i$ the value predicted by the model, and $n$ the total number of samples. By restoring the original scale of the target variable, RMSE keeps the heavy penalty on large errors imposed by MSE and permits intuitive, scale-aware comparisons across models or datasets.

(5) Mean absolute error (MAE)

Mean absolute error quantifies the average magnitude of absolute deviations between predictions and observations. Its formula reads

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

with $y_i$ the true value of the $i$-th sample, $\hat{y}_i$ the corresponding prediction, and $n$ the sample size. Because every residual contributes linearly, MAE is less sensitive to outliers than squared-error metrics and offers an immediately interpretable statement of "how far off" the model is on average.

(6) Coefficient of determination (R2)

R2, or the coefficient of determination, measures the proportion of variance in the dependent variable that is predictable from the independent variables. It is computed as

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \overline{y}_i)^2}$$

where $y_i$ is the true value of the $i$-th sample, $\overline{y}$ the mean of all true values, $\hat{y}_i$ the model's prediction, and $n$ the number of samples. A value of 1 indicates perfect explanatory power, 0 shows the model performs no better than the mean baseline, and negative values reveal performance worse than simply predicting the mean.

(7) Area under the ROC curve (AUC)

AUC summarises a binary classifier's ability to rank positive instances ahead of negative ones. It equals the area under the receiver operating characteristic curve, whose abscissa is the false positive rate (FPR) and whose ordinate is the true positive rate (TPR). Formally,

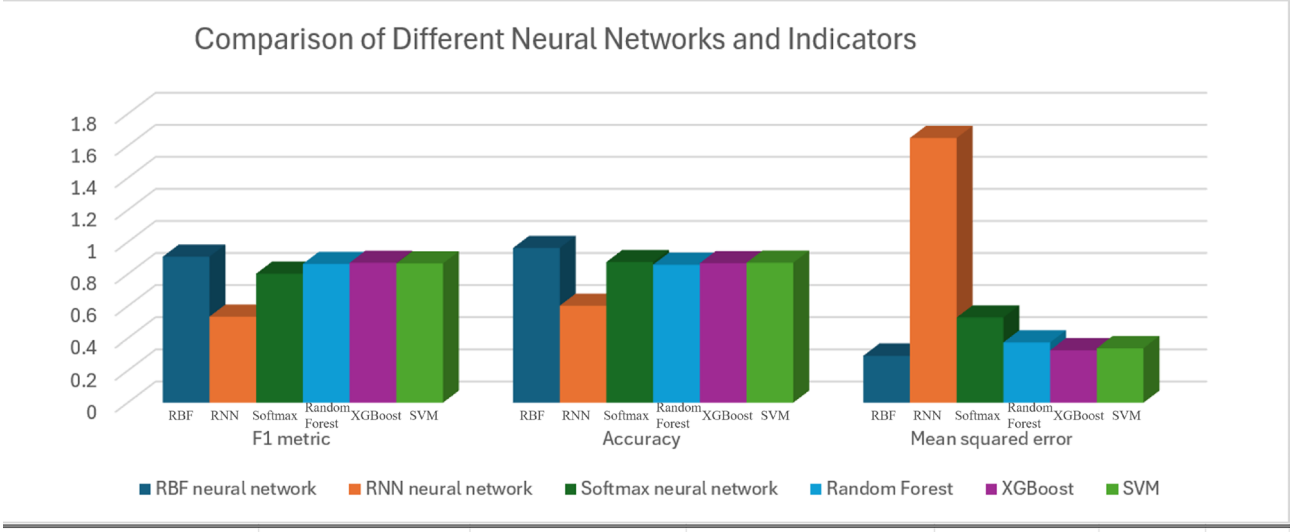
$$\mathrm{AUC} = \int_0^1 TPR(FPR^{-1}(u))du$$

AUC ranges from 0 to 1. A score of 0.5 corresponds to random ranking; higher values denote superior discrimination, whereas values below 0.5 indicate performance worse than chance, but still interpretable after reversing the predicted scores.

(8) Precision-Recall Curve and average Precision (AP)

| Indicator\network type | RBF neural network | RNN neural network | Softmax neural network | Random forest | XGboost | SVM |
|---|---|---|---|---|---|---|
| F1 metric | 0.909 | 0.535 | 0.803 | 0.864 | 0.871 | 0.867 |
| Accuracy | 96.32% | 60.30% | 87.60% | 0.859 | 0.869 | 0.8714 |
| Mean squared error | 0.2926 | 1.65 | 0.53 | 0.3741 | 0.3263 | 0.339 |

**Table 10**. Performance comparison results of different models.




| Indicator/Network Type | RBF neural network | RNN neural network | Softmax neural network | Random Forest | XGBoost | SVM |
|---|---|---|---|---|---|---|
| F1 metric | 0.909 | 0.535 | 0.803 | 0.864 | 0.871 | 0.867 |
| Accuracy | 96.32% | 60.30% | 87.60% | 0.859 | 0.869 | 0.8714 |
| Mean squared error | 0.2926 | 1.65 | 0.53 | 0.3741 | 0.3263 | 0.339 |

**Fig. 12**. Comparison of different neural networks and indicators.

The Precision-Recall curve plots Precision against Recall while sweeping the decision threshold. Average Precision is the area under this curve and can be approximated by

$$\mathrm{AP} = \sum_{k=1}^{K} (R_k - R_{k-1}) P_k$$

where $P_k$ is the precision observed at the $k$-th threshold and $R_k - R_{k-1}$ the corresponding increase in recall. AP provides a single scalar that summarises performance across all operating points; it is especially informative when the positive class is rare, because it focuses exclusively on the behaviour of the classifier among the actual positives and therefore complements the ROC-based AUC in imbalanced scenarios.

These metrics can quantify the disparity between predicted values and actual values from different models, thereby offering an objective basis for comparing model performance. The comparative results of different models' performances are summarized in Table 10.

A comparison of different neural networks and metrics is shown in Fig. 12:

The analysis results show that the RBF neural network achieves 0.909 in F1 metric, 96.32% accuracy, and 0.2926 mean square error, which are better than the RNN and Softmax regression models. The RNN model has an F1 metric of 0.535, an accuracy of 60.3%, and a mean square error of 1.65, which indicates that its performance is relatively weak. The Softmax regression model has a good F1 measure of 0.803, but its accuracy of 87.6% and mean square error of 0.53 are still not as good as the RBF neural network. Random Forest, XGBoost and SVM form a highly competitive second tier: their F1-scores fall within 0.864–0.871, accuracies cluster around 86–87% and MSEs hover between 0.326 and 0.374, yet none surpasses the RBF.

In order to comprehensively assess and compare the performance of these models, the cross-entropy loss function is also utilized as the optimization criterion. The loss function curves of different models are shown in Fig. 13.

In this study, RBF neural network demonstrated significant superiority in the academic warning task. After 100 rounds of iterations, the loss function of the RBF neural network decreased significantly and stabilized. It

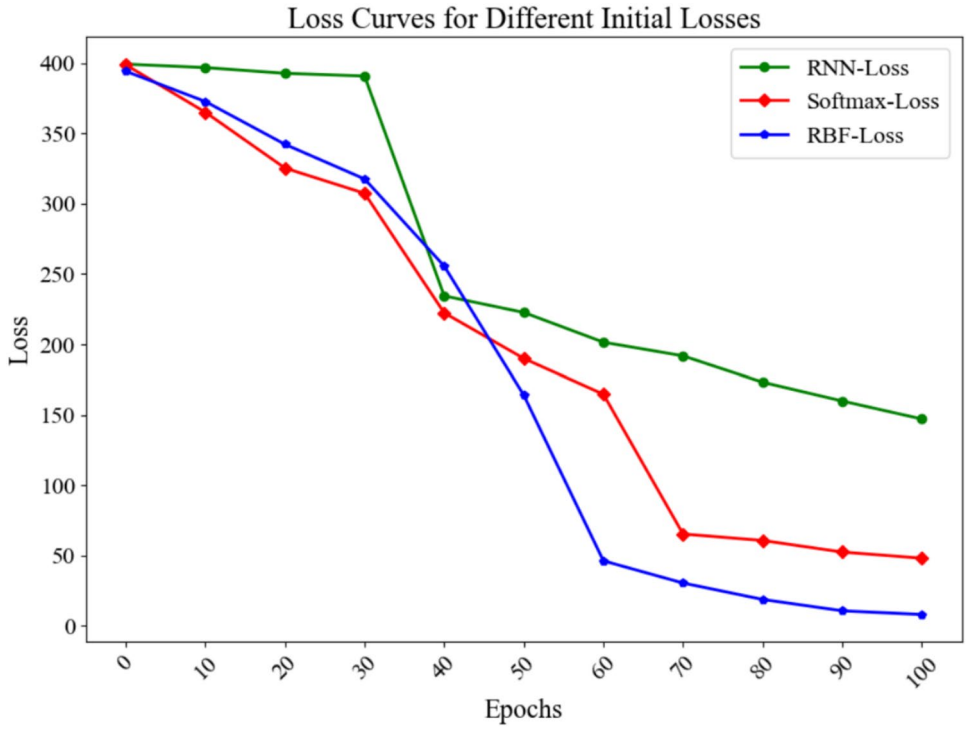



**Fig. 13**. Loss function curves of different models.

indicates the model has learned the intrinsic patterns of the data quickly and has effective prediction capabilities. In contrast, the recurrent neural network (RNN) and Softmax regression models show a slower decrease in loss and fail to reach the performance level of the RBF neural network with the same number of iterations. This phenomenon implies that the RNN and Softmax regression models may require a greater number of iterations to achieve comparable performance to RBF neural networks.

To further demonstrate the model's performance and generalization capability, we present the AUC curve, Precision–Recall curve, and tenfold cross-validation summary plot in Fig. 14.

The figure result consolidates three complementary views of model quality. Fig(a) the tenfold CV performance profile shows tight distributions of RMSE (mean 0.18), MAE (mean 0.15) and R2 (mean 0.74), attesting to stable generalisation. Fig(b) the ROC curve yields an AUC of 0.96, signalling excellent separation between classes. Fig(c) the Precision–Recall curve delivers an AP of 0.96, corroborating strong reliability even under pronounced class imbalance.

In summary, RBF neural networks demonstrate their superiority in the academic warning task in terms of loss function descent rate, number of iterations, and prediction accuracy.

## Model application

Through the previous summaries of the system's requirements analysis and system functional module design, this chapter summarizes the implementation of the entire academic early warning system, introduces the system's development environment and detailed data mining module implementation, system management module implementation, and system setup module implementation. The total architecture of the academic early warning system is shown in Fig. 15:

## Introduction to the experimental environment

This system is mainly composed of two parts: data mining and system management, in which the information of the software environment used is shown in the Table 11:

The hardware environment is mainly the computers equipped in the school as shown in the Table 12.

## Management module implementation

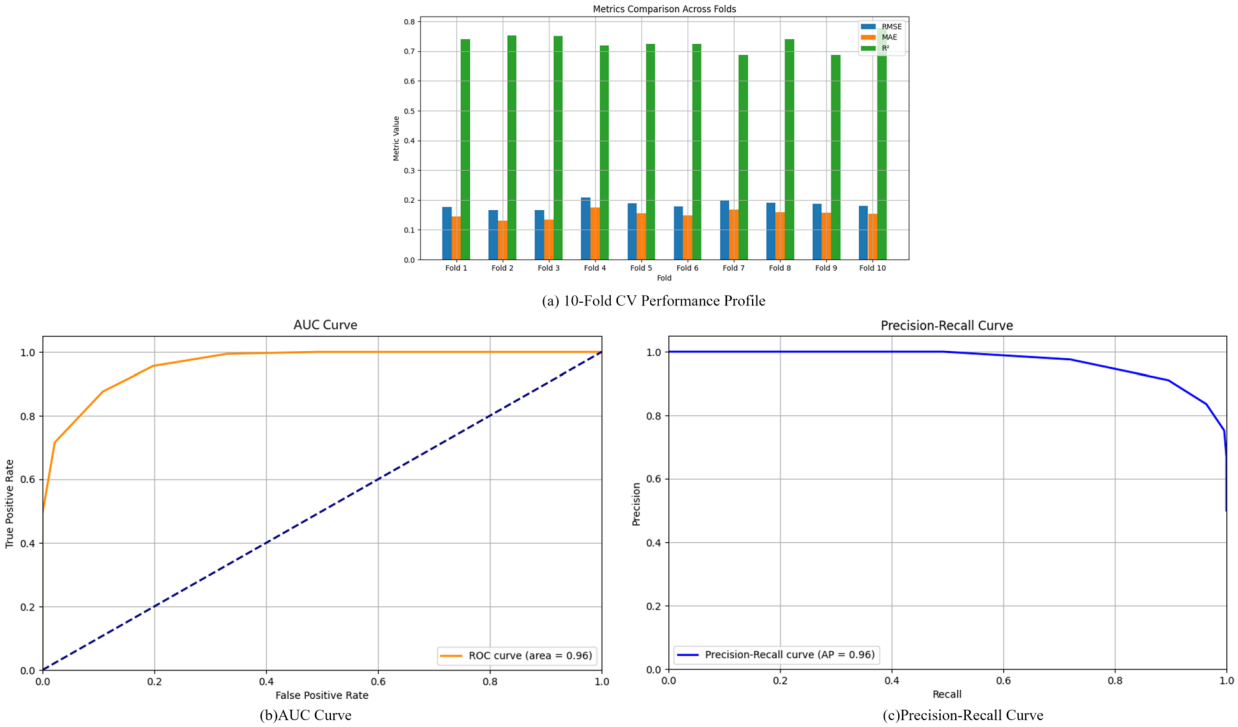
The early warning management module mainly consists of three sub-modules: user login, importing student data, exporting student data, and exporting early warning list. Academic early warning system administrator and teacher user login interface where the account password is added by the administrator background and saved in the database, the login page is shown in Fig. 16 Student user login interface, the account for the student's student number, the password for the student's ID card number after six, the login interface.

(1) Importing the student data page shown in Fig. 17.

(2) Exporting student data.

Then click the "Save" button and export the student data as shown in Fig. 18.

(3) Export of the early warning list.




(a) 10-Fold CV Performance Profile

(b)AUC Curve

(c)Precision-Recall Curve

**Fig. 14**. Comprehensive tenfold cross-validation assessment of the proposed model, AUC curve and Precision-Recall curve.

After the academic warning list, click the "Export" button to export the academic warning list, as shown in Fig. 19.

## Conclusions

In this study, we have successfully developed and implemented a dynamic academic early warning system based on radial basis function (RBF) neural networks. The system provides a scientific and efficient prediction and intervention tool for the current crises that students may encounter in the academic process. Through in-depth research on the completeness of the academic early warning indicator system, the completeness of the model, the diversity of the calculation methods, and the advanced technical means, this system shows significant advantages in several aspects:

(1) Improvement of the system of academic early warning indicators:

Through the FCE-AHP method of screening, 10 key academic influences were identified from 30 preliminary indicators, and the weights of these factors were distributed as follows: classroom attendance (8.526%), implementation of personal study plan (11.126%), participation in disciplinary competitions (9.375%), etc., and each of these factors passed the scoring and consistency test by the experts (with a CR value of 0.093) , which ensures the scientificity and rationality of the indicator system.

(2) Completeness of academic early warning models:

The designed RBF neural network model showed high prediction accuracy in academic crisis analysis. The model showed a significant decrease in the loss function after 100 iterations, demonstrating the ability to learn quickly and predict academic crises effectively. Compared with RNN and Softmax regression models, the RBF model is more advantageous in terms of loss function decline speed and prediction accuracy.

(3) Diversity of methods for calculating academic alerts:

The system employs a variety of data preprocessing techniques and machine learning algorithms to improve the diversity and accuracy of early warning calculations. Data preprocessing includes normalization processing, which ensures the generalization ability of the model. SGD is used for model training, which improves the optimization efficiency of model parameters through iterative optimization.

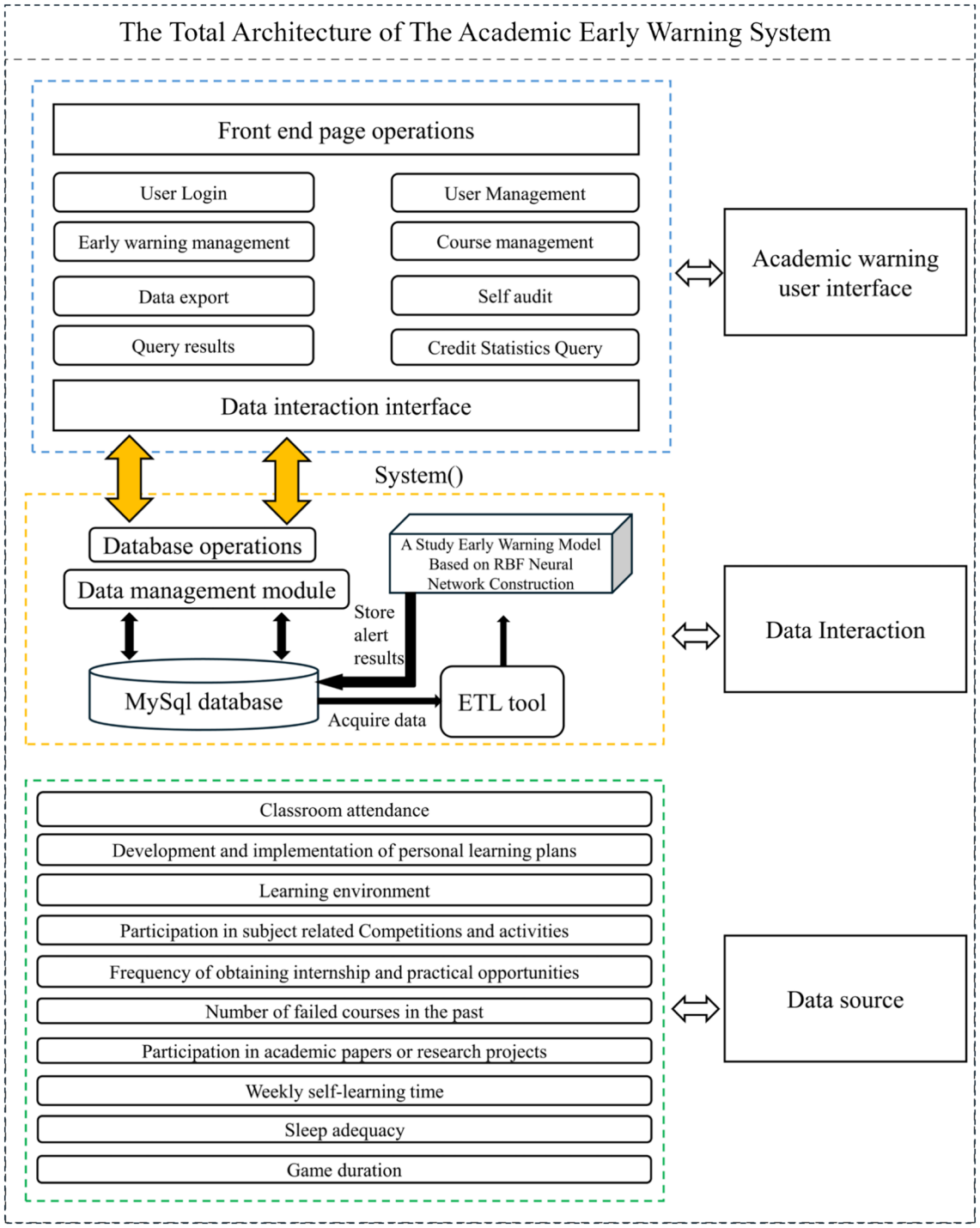



**Fig. 15**. The total architecture of the academic early warning system.

| Environmental description | Parameter |
|---|---|
| Development language | HTML, css, JavaScript, Python |
| Development tool | Vscode |
| Database | MySQL |
| Code management tools | Git |

**Table 11**. Software environment information.

| Environmental description | Parameter |
|---|---|
| Operating system | Ubuntu 22.04 |
| Processing unit | Intel i9-11900 K |
| Motherboard (computer) (lit. lord board) | Prime Z790-P |
| Random access memory (RAM) | f5-6400j3239g16gx2-tz5rk(32 g) |
| Hard disk | Samsung SSD 970 EVO Plus 1 TB |
| Display card (computer) | RTX3080 |

**Table 12**. Hardware environment information.

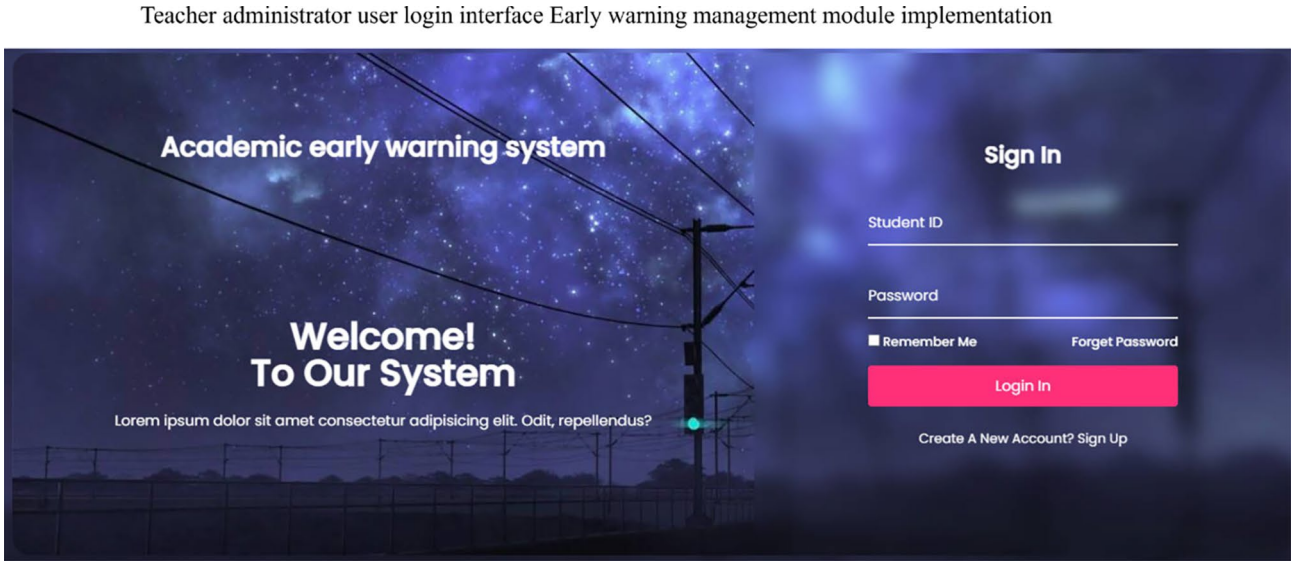



**Fig. 16**. Teacher administrator user login interface early warning management module implementation.

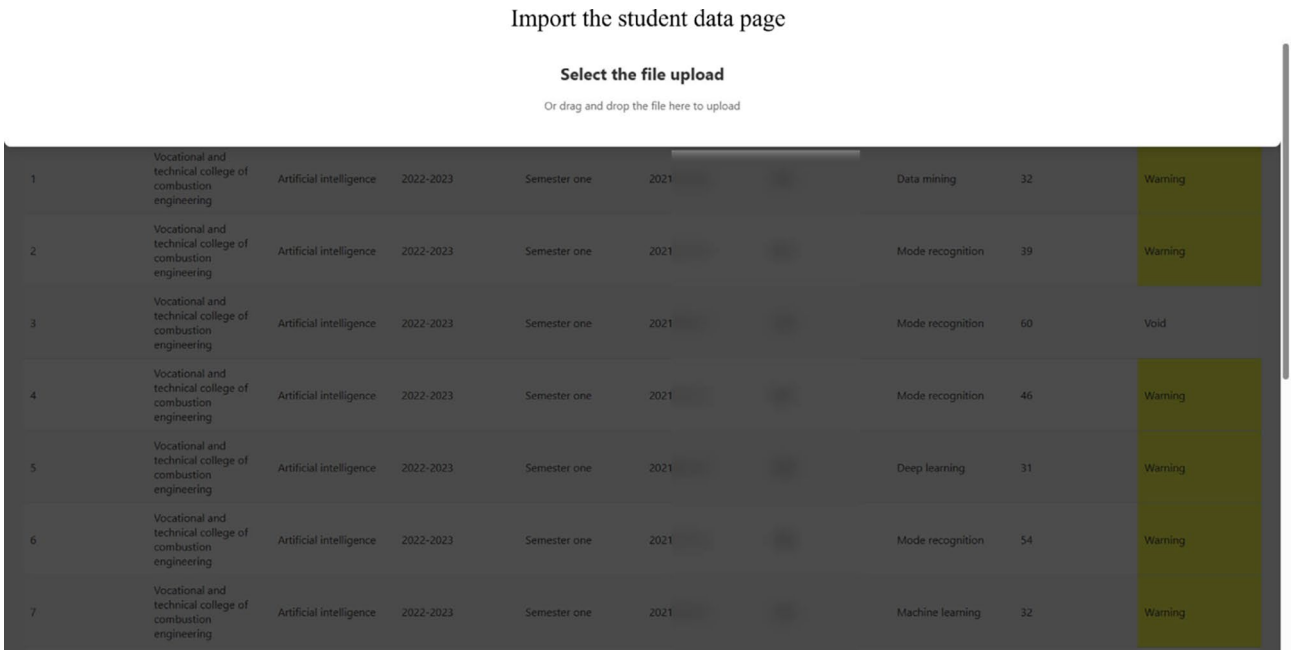



**Fig. 17**. Import the student data page.

Student data export interface

College: Vocational and technical co | Profession: Artificial intelligence | Academic Year: 2022-2023 | Grade: 70 and below | Inquire | Reset

| Serial Number | College | Profession | Academic Year | Semester | Student ID | Name | Course Title | Grade |
|---|---|---|---|---|---|---|---|---|
| 1 | Vocational and technical college of combustion engineering | Artificial intelligence | 2022-2023 | Semester one | 2021 | | Data mining | 64 |
| 2 | Vocational and technical college of combustion engineering | Artificial intelligence | 2022-2023 | Semester one | 2021 | | Mode recognition | 61 |
| 3 | Vocational and technical college of combustion engineering | Artificial intelligence | 2022-2023 | Semester one | 2021 | | Mode recognition | 58 |
| 4 | Vocational and technical college of combustion engineering | Artificial intelligence | 2022-2023 | Semester one | 2021 | | Machine learning | 49 |
| 5 | Vocational and technical college of combustion engineering | Artificial intelligence | 2022-2023 | Semester one | 2021 | | Data mining | 51 |
| 6 | Vocational and technical college of combustion engineering | Artificial intelligence | 2022-2023 | Semester one | 2021 | | Mode recognition | 59 |
| 7 | Vocational and technical college of combustion engineering | Artificial intelligence | 2022-2023 | Semester one | 2021 | | Introduction to Artificial Intelligence | 37 |

**Fig. 18**. Student data export interface.

Export interface for academic warning list

College: Vocational and technical co | Profession: Artificial intelligence | Academic Year: 2022-2023 | Grade: 70 and below | Inquire | Reset

| Serial Number | College | Profession | Academic Year | Semester | Student ID | Name | Course Title | Grade | Forewarning |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Vocational and technical college of combustion engineering | Artificial intelligence | 2022-2023 | Semester one | 2021 | | Introduction to Artificial Intelligence | 44 | Warning |
| 2 | Vocational and technical college of combustion engineering | Artificial intelligence | 2022-2023 | Semester one | 2021 | | Data mining | 51 | Warning |
| 3 | Vocational and technical college of combustion engineering | Artificial intelligence | 2022-2023 | Semester one | 2021 | | Machine learning | 38 | Warning |
| 4 | Vocational and technical college of combustion engineering | Artificial intelligence | 2022-2023 | Semester one | 2021 | | Introduction to Artificial Intelligence | 69 | Void |
| 5 | Vocational and technical college of combustion engineering | Artificial intelligence | 2022-2023 | Semester one | 2021 | | Deep learning | 58 | Warning |
| 6 | Vocational and technical college of combustion engineering | Artificial intelligence | 2022-2023 | Semester one | 2021 | | Mode recognition | 41 | Warning |
| 7 | Vocational and technical college of combustion engineering | Artificial intelligence | 2022-2023 | Semester one | 2021 | | Machine learning | 65 | Void |

**Fig. 19**. Export interface for academic warning list.

(4) The sophistication of the technical means of academic early warning:

The implemented academic alert system provides real-time warning alerts and is capable of handling large-scale student data (e.g., a dataset of 1100 students). The results of the system test show that the average absolute error between the predicted and actual values is 0.2405, and the root mean square error is 0.2926, with an accuracy of 90.5%, which proves the system's high sensitivity and ability to recognize correctly.

This study also has several limitations. For instance, the sample was confined to engineering students and comprised a high proportion of males, which may limit the generalizability of the findings.

In summary, the academic early warning system based on RBF neural network developed in this study has significant technical advantages in providing personalized academic support and real-time early warning, which can satisfy the demand for academic early warning tools in the field of higher education, and is of great practical value in enhancing students' academic success.

## Method

1. We confirm that all methods were carried out in accordance with the relevant guidelines and regulations.
2. The study was reviewed and approved by the College of Engineering, Hebei Normal University.
3. Written informed consent was obtained from all participants (or their legal guardians) prior to participation in the survey.

## Data availability

The data used in this paper is available from the corresponding author.

## Code availability

To ensure long-term accessibility and reproducibility, a version-controlled archive of the code has been created and is available via Zenodo: https://doi.org/10.5281/zenodo.19723846

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## Author contributions
Yihan Liu: Writing—review and editing, writing—original draft, investigation, conceptualization. Rui Ma: Writing—review and editing, project administration, methodology, data curation. Qike Wu: Writing—review and editing, writing—original draft, investigation, conceptualization. Qiang Li: writing—review and editing, writing—original draft, investigation, conceptualization.

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## Declarations

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Participants consented to having their data used after anonymization.

## Additional information
**Correspondence** and requests for materials should be addressed to Y.L.

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